Demystifying ehancer-gene linkage using population-scale single-cell multiomics

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Abstract

The epigenetic regulation of gene expression is highly cell-type and context specific. Recent advances in single-cell RNA and ATAC multiomic technologies provide unprecedented opportunities to map enhancers to their target genes. Existing methods, however, largely rely on enhancer-gene correlation either across individual pseudobulks or across heterogeneous cell types. We develop a Mendelian randomization framework leveraging population-scale single-cell multiomics to directly identify causal and pleiotropic enhancer-gene relationships, one cell type at a time. We apply our method to single-cell RNA and ATAC data of 428 individuals from the Chinese Immune Multiomics Atlas and show that the resulting enhancer-gene regulatory landscape differs substantially from those identified by existing approaches. We orthogonally validate our findings using promoter capture Hi-C and H3K27ac HiChIP data and demonstrate significant enrichment of both transcription factor motifs and blood-trait heritability among the identified enhancer-gene pairs.

Author Summary

Introduction

Results

Study overview and comparison of peak–gene linking methods

In the study, we used three common causal methods (IVW, MR-Egger, SMR) (reference) and two association based methods to study peak-gene pairs. We use the terms: causal IVW, causal MR-Egger, and causal SMR to refer the three causal methods throughout the paper. The six methods were three basic MR methods (causal IVW, causal MR-Egger, causal SMR), an equal-weight Cauchy combination of these three MR tests (causal joint), WNN metacell Spearman correlation (marginal multiomics), and donor-level pseudobulk Spearman correlation (marginal pseudobulk).

First, we implemented MR methods to study the CIMA project data, which was a comprehensice multi-omics analysis of 428 healthy Chinese adults, incorporting scRNA-seq and scATAC-seq of circulating immune cells along with whole-genome sequencing (WGS) of plasma cell-free DNA (reference). CIMA paper applied TensorQTL to identify cell-type specific cis-xQTLs within -+ 1Mb of gene transcription start sites (TSSs) or the chromatin peak midpoints. We directly used cell-type-specific caQTL and eQTL summary statistics to investigate accessibility-to-expression links in naive B cells (Bn), naive $CD4^+$ T cells (CD4 T cells), and naive $CD8^+$ T cells (CD8 T cells). Second, for the marginal pseudobulk method, we did quality control and selected passed individuals on the CIMA scRNA and scATAC counts (see methods), and then we aggregate the gene expression or chromatin accessibility for each select individuals and for each cell types (335 individuals for Bn, 372 for CD4 T cell, and 346 for CD8 T cell). Spearman correlation was calculated between the pseudobulk gene expression and chromatin accessibility across individuals. Third, for the marginal multiomics, we used an external multiomics PBMC data. We aggregated the gene expression or chromatin accessibility for each meta cells (80 meta cells in total) from WNN methods, and then calculate the Spearman correlation across metacells. (Fig 1A)

All six methods are applied to study the same set of candidate peak-gene pairs (40,087 for Bn, 85,527 for CD4 T cell, and 50,931 for CD8 T cell) for fair comparison (reference, supp table 3). Marginal multiomics method found 37,181 significant peak-gene pairs (BH p.adjusr < 0.1) for CD4 T cells, which was much more than other methods. Causal methods (causal IVW, causal MR Egger, and causal SMR) generally found less significant peak-gene links than the two marginal association methods (marginal multiomics and marginal pseudobulk), which was reasonable, since the causal methods focus on detecting the causal relationship between peaks and genes. The set of significant peak-gene pairs are overlapped between different methods,

which might show the consistency across different methods. (Figure 1B). The results of Bn and CD 8 T cells is samilair. (supplementary figure 3).

Even if some peak-gene pairs are significant in both marginal and causal methods, the directions might be different. We extract the all 1259 peak-gene pairs which were significant for all six methods for CD4 T cells, and check the effect direction. The results of causal IVW, causal MR-Egger, and causal SMR were highly concordant. There were 73% of significant pairs having the the same sign between causal IVW and the two marginal methods, while 24% of significant pairs results were not concordant between causal IVW and marginal multiomics. We found the marginal multiomics methods was more likely to be discordant with causal IVW than the marginal pseudobulk method (Figure 1C, Figrue S3).

Concordance and discordance between MR estimates and marginal correlations

Mendelian randomization tries to find the causal regulation effect on accessibility peaks on the gene expression, while the marginal multiomics and marginal pseudobulk methods calculate the correlation between peaks and genes across cell types and individuals separately. The causal effects and marginal associations agree on many peak-gene pairs, which is reasonable. But discordance still exsits. We select four representative case to illustrate this.

For the gene SIDT2 and the peak chr11:117185715-117186216, the MR results (causal SMR, causal IVW, causal Egger, and causal joint) and two marginal correlations (marginal multiomics and marginal pseudobulk) were all positive and significant (BH p.adjast < 0.1)(Figure 2A). Causal IVW and causal Egger had the most significant results, and the two lines were closely aligned implying no clear pleiotropy in the data (Figure 2A). However, the concordance of significance did not always exist among methods. For second examples (gene RPL11, peak chr1:23691494-23691995) in CD4 T cell, while the MR estimates and the correlation across metacells are significant, the correlation across individuals was not significant anymore (Figure 2B). Therefore, the causal effects were not always mirrored by a marginal association within the same cell type.

The causal effects and the marginal associations might have even different sign. For the gene ERAP2 and the peak chr5:96885894-96886395 in CD4 T cell, all of causal IVW, causal Egger, and causal SMR detect positive and significant effects, consistent with the negative correlation across individuals ($\rho = -0.641$) (Figure 2C). However, the correlation across cell types was surprisingly positive ($\rho = 0.375$, p= $6.1e-4$), indicating the possible directional discrepancy between causal effects and marginal association.

MR methods could unique causal effects which were possibly ignored by marginal correlation. For example, all MR methods detect a significantly positive causal effect of the peak ARPC1B on the peak chr7:99376204-99376705 in CD4 T cell, whereas neither marginal correlation tests provided evidence of the

association. (Figure 2D)

Together, these examples show the consistency and possible difference between marginal correlation and causal effects. Since the marginal multiomics and marginal pseudobulk only calculated the marginal association, which might be influenced by potential unobserved counfunding factors. The calculated correlation may be different or even of different signs with the causal effects. This demonstrates that when studying the regulation mechanism between peak accessibility and gene expression, we should not only study the marginal association, but also find the underlying causal effects.

Functional characterization and regulatory network support for peak–gene links

To characterize the biological functions of the results MR methods, we performed the GO enrichment analysis using DAVID website (reference) on the genes only implicated by MR methods. We collected all peak-gene pairs which were significant by at least one MR method (causal IVW, causal Egger, and causal SMR p adjust < 0.1), but were only significant in both marginal association (marginal multiomics, marginal pseudobulk P.adjust > 0.1) from the three cell types. We extract all genes (4,031 unique genes) for the following GO enrichment analysis. DNA-templated transcription was the most significant biological process term, including 689 genes, with a fold enrichment of 1.36 and a DAVID-reported FDR of 5.18×10^{-18} (Figure 3A). The molecular function enrichment showed protein binding (fold enrichment 1.16, DAVID FDR 2.81×10^{-85}), RNA binding (fold enrichment 1.41, DAVID FDR 3.94×10^{-15}), and zinc ion bind terms (fold enrichment 1.33, DAVID FDR 3.58×10^{-15}) are most significant terms (Figure 3B). Genes related with the RNA-binding enrichment, such as SRSF7 amd DDX5, were known to have establilshed role in pre-mRNA splicing and RNA processing (reference). Genes included in the zinc binding terms, such as ZNF143 and IKZF3, showed the potential presence of zinc-finger regulatory proteins (reference).

By integrating scRNA-seq and scATAC-seq data, CIMA applied SCENIC+ (version 1.0.1) to construct cell-type specific enhancer-linked regulatory units (eRegulons) and further select a subset of high-quality eRegulons. Each eRegulons is a triple of (TF, chromatin peak, gene), which can reasonably become a validation data. We extract unique region-gene pairs from the CIMA eRegulons and calculate the adds ratio with the significant peak-gene links identifies by each method(Figure 3c). Although marginal multiomics and marginal pseudibulk method tended to find more significant peak-gene pairs (Figure 1c, Figure S3), causal methods showed more significant enrichment in network-spported links across cell types, especially among the high-quality eRegulon units. Although overlapping with CIMA regulatory network were not the perfect ground-truth of causal regulation effects, it could partially support the reliability of MR estimates.

Disease heritability enrichment and GWAS overlap of MR-supported peak–gene links

To explore the disease relevance of the peak-gene pairs found by MR pairs, we assessed enrichment scores using stratified LD score regression. Specifically, we extracted all peak-gene pairs which were significant in at least one causal methods (causal SMR, causal Egger, and causal IVW) but not significant in both marginal association methods. We pooled the peaks from all selected peak-gene pairs from three cell types, and got 3,777 unique peaks (MR Only). We constructed the annotation based on the selected 3,777 peaks and run S-LDSC model adjusting 97 baselineLD annotations (v2.2). Besides, background peaks (All peaks) from all available cis-QTL results were also analyzed by the same S-LDSC analysis pipeline for comparison (Figure 4A). We select the seven GWAS triated studied in OneK1K paper (OneK1K reference). Enrichment scores were defined as the ratio of heritability explained to the proportions of variants. MR-only peaks had higher enrichment point estimates than all candidates peaks across seven GWAS traits (Figure 4A). Nominal evidence of enrichment at $P < 0.1$ was observed for Crohn's diseases, inflammatory bowel disease, multiple sclerosis, and type 1 diabetes. (Figure 4A). By the results of S-LDSC, we could say that peaks only detected by MR methods had enriched signals of some immune diseases, implicating some biological process only detected by MR. It demostrated the importance of using MR to study the causal relation between peaks and genes.

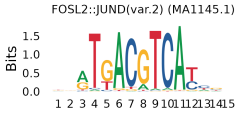
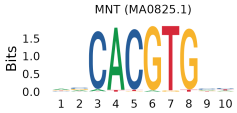
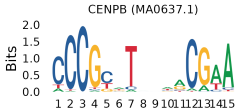
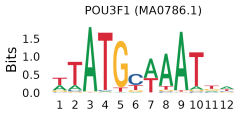
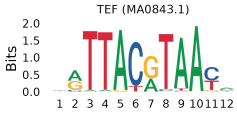
We next showed individual MR-supported links whose peak regions contained genome-wide significant variants. Those links were detected by at least one of causal methods, whereas neither marginal correlation analysis reached significance (p adjust < 0.1). For examples, an MR-supported peak linked to the gene ERBB3 contained rs705704, a variants significantly associated with type 1 diabetes. This finding nominated ERBB3 as a candidate target gene of the GWAS-overlapping regulatory in CD4 T cells. Additional examples included RDL2-linked peaks overlapping type 1 diabetes and rheumatoid arthritis variants, a POLR2E-linked peak overlapping Crohn's disease variants, and a UBAC2-linked peak overlapping asthma variants. Although we cannot directly say there were causal relationship between the genes and GWAS diseases, these MR-supported peak-gene links provided a potential regulation mechenism ways for these diseases.

Motif enrichment highlights candidate regulators of MR-supported peak–gene links

To characterize the regulatory sequence features of peaks by MR methods, we performed transcription factor motif enrichment analysis using all available peaks as background matched for GC content and chromatin accessibility. We focused on the all unique peaks from the peak-gene pairs which were significant in at least

one of causal methods but were not significant under both of two marginal correlation methods. We show the most significant Motif enrichment results for each cell types (Table 1). In CD4 T cells, the MNT motif had the strongest enrichment (p adjust = 0.003, XXX fold enrichment for causal Egger), which were related with the regulation of T-cell proliferation and immune homeostasis by the previous study (reference). In Bn cells, causal Egger-selected peaks were shown the enrichment in the FOSL2::JUND motid, which were one of AP-1 family members and were known to regulate B-cell proliteration and pre-B-cell differentiation through IRF4 and Foxol. Other selected motifs included ATF3 in Bn cells, CENPB in CD4 T cells, and POU3F1 and TEF in CD8 T cells.

Table 1. MR-specific transcription factor motifs in peaks supported by Mendelian randomization.

Motif logo	p.adj IVW	p.adj MR-Egger	p.adj SMR	p.adj CCT	p.adj WNN	p.adj Pseudo-bulk	Brief interpretation & link
Bn							
	**0.041**	**1.5e−4**	0.157	0.059	0.072	0.056	In mouse marginal zone B cells, ATF3 drives PDL1 expression, through which these cells restrain the T follicular helper response to a high-cholesterol diet. https://pubmed.ncbi.nlm.nih.gov/28414328/
	0.057	**0.001**	0.137	0.056	0.088	0.059	Fra-2 (FOSL2) drives IRF4 and Foxo1 transcription and is required for mouse B-cell proliferation and pre-B-cell differentiation. https://pubmed.ncbi.nlm.nih.gov/28566276/
CD4 T cell							
	0.176	**0.003**	0.051	**0.032**	0.077	0.113	Deleting Mnt in mouse T cells causes Th1-skewed hyperproliferation of activated CD4$^+$ T cells, intestinal inflammation and T-cell lymphoma. https://pubmed.ncbi.nlm.nih.gov/16507988/
	**0.035**	**0.004**	0.172	**0.033**	0.329	0.156	CENP-B is a major anti-centromere autoantigen; antibodies to recombinant CENP-B were detected in 60% of patients with primary biliary cholangitis. https://pubmed.ncbi.nlm.nih.gov/8527711/
CD8 T cell							
	0.415	**0.006**	0.591	0.140	0.499	0.232	POU3F1 is one of the top five transcription factors predicted from disease-associated chromatin accessibility and gene expression in CD4$^+$ T cells of SLE patients. https://pubmed.ncbi.nlm.nih.gov/36738683/
	0.112	**0.010**	0.067	0.124	0.117	0.087	TEF is expressed in hematolymphoid progenitors, where it blocks apoptosis on IL-3 withdrawal and restrains proliferation by down-regulating the common beta chain of cytokine receptors. https://pubmed.ncbi.nlm.nih.gov/15665112/

Motif enrichment was tested for peaks selected by each method (within-gene BH-adjusted $q < 0.1$) against a background matched on GC content and accessibility (Signac, JASPAR2020 vertebrate CORE motifs). MR-specific motifs have p.adj < 0.05 and fold enrichment > 1 in at least one of IVW, MR-Egger, and SMR, with no significant enrichment (p.adj ≥ 0.05) in WNN or Pseudobulk; for each cell type, the two with the smallest MR p.adj and a clear literature-based interpretation are shown. Bold, p.adj < 0.05 with fold enrichment > 1. p.adj is the adjusted motif-enrichment p-value reported by Signac for each method. Each interpretation is based on the single paper linked below it, checked against its PubMed record. Pseudobulk, pseudobulk correlation; WNN, WNN metacell correlation; CCT, Cauchy combination of SMR, IVW, and MR-Egger.

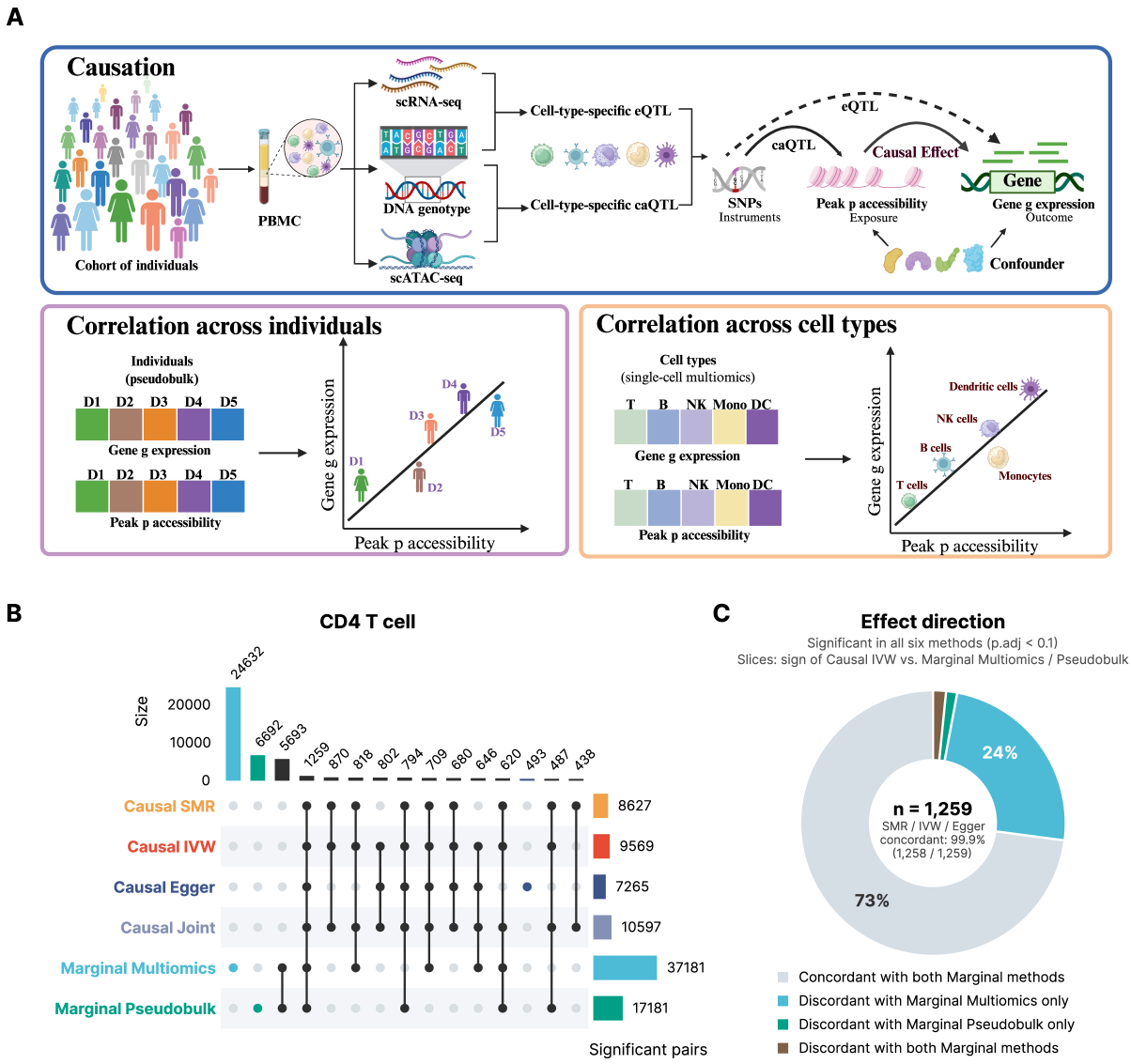


Fig 1. The complete analysis framework and the results of CD4 T cells. **A** MR methods (IVW, MR Egger, SMR) use cell-type specific caQTL/eQTL summary statistics; Marginal multiomics uses external PBMC metacells, and marginal pseudobulk correlations use CIMA individuals. Both two marginal correlation are measured by the Spearman correlation. **B** UpSet plot of significant links in CD4 T cells (p-adjust < 0.1, BH) **C** Effect directions among the 1,259 CD4 T cell links significant in all six methods. Slices compare the sign of the IVW estimate with the signs of the marginal correlations: concordant with both marginal methods (light gray), discordant with Marginal Multiomics only (blue), discordant with Marginal Pseudobulk only (green), and discordant with both (brown). Bn and CD8 T cell comparisons appear in S3 Fig.

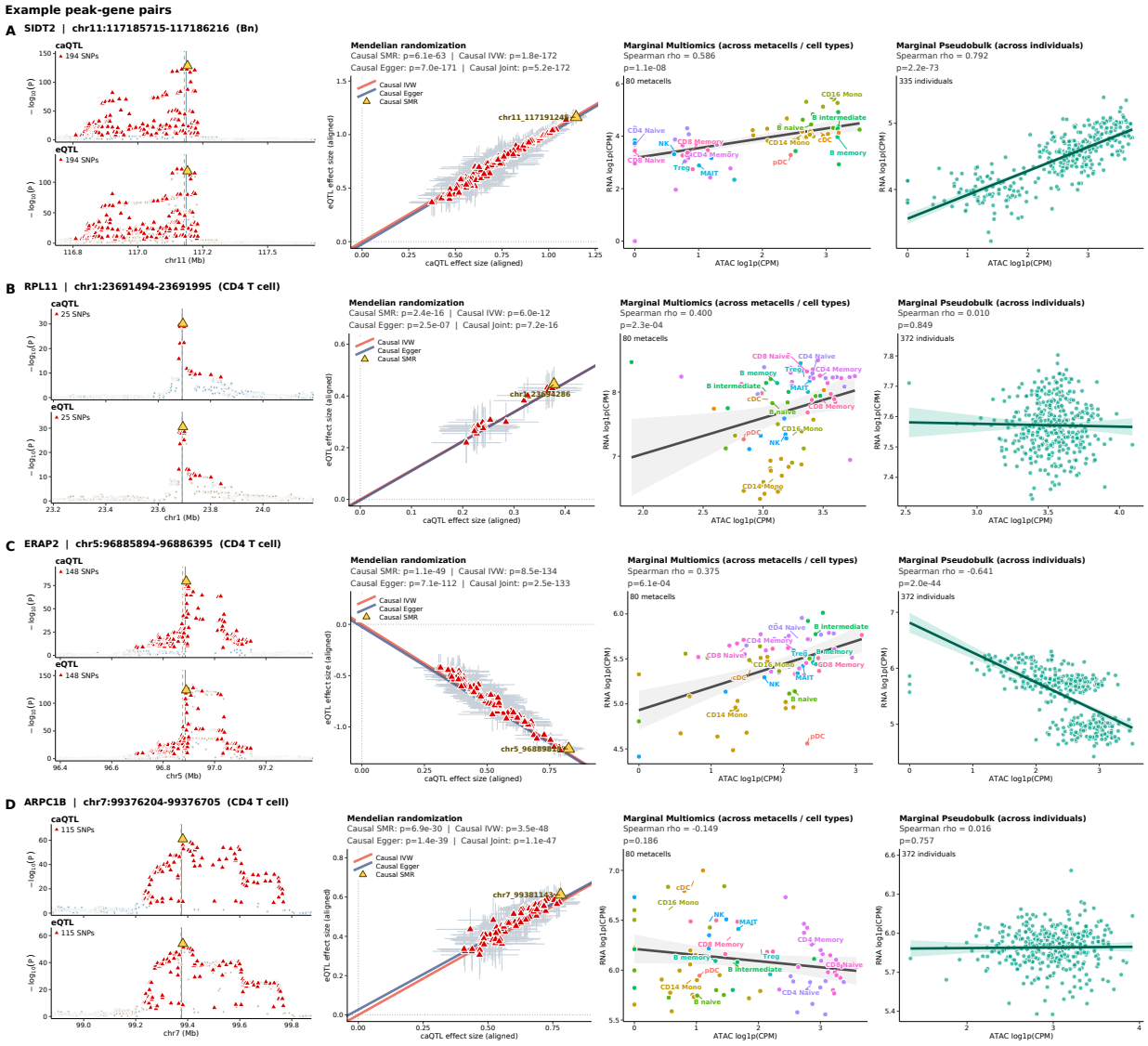


Fig 2. Four selected peak–gene links. Rows show four peak–gene pairs **First column:** Regional caQTL (top) and eQTL (bottom) tracks. The horizontal axis indicates genomic position, and the vertical axis shows $-\log_{10}(p)$. Red triangles mark retained MR instruments, and yellow triangles identify the SNP selected for SMR. **Second column:** MR method results, red and blue lines show the IVW and MR-Egger fits, respectively. The yellow triangle highlights the leading SNP used in SMR. Report p-values correspond to SMR, IVW, MR-Egger, and the Cauchy combination ("Causal Joint"). **Third column:** Peak accessibility and gene expression across 80 WNN metacells from an external PBMC multiome dataset. Each point represents a metacell, colored by cell type **Fourth column:** Peak accessibility and gene expression across matched individuals. Each point represents an individual.

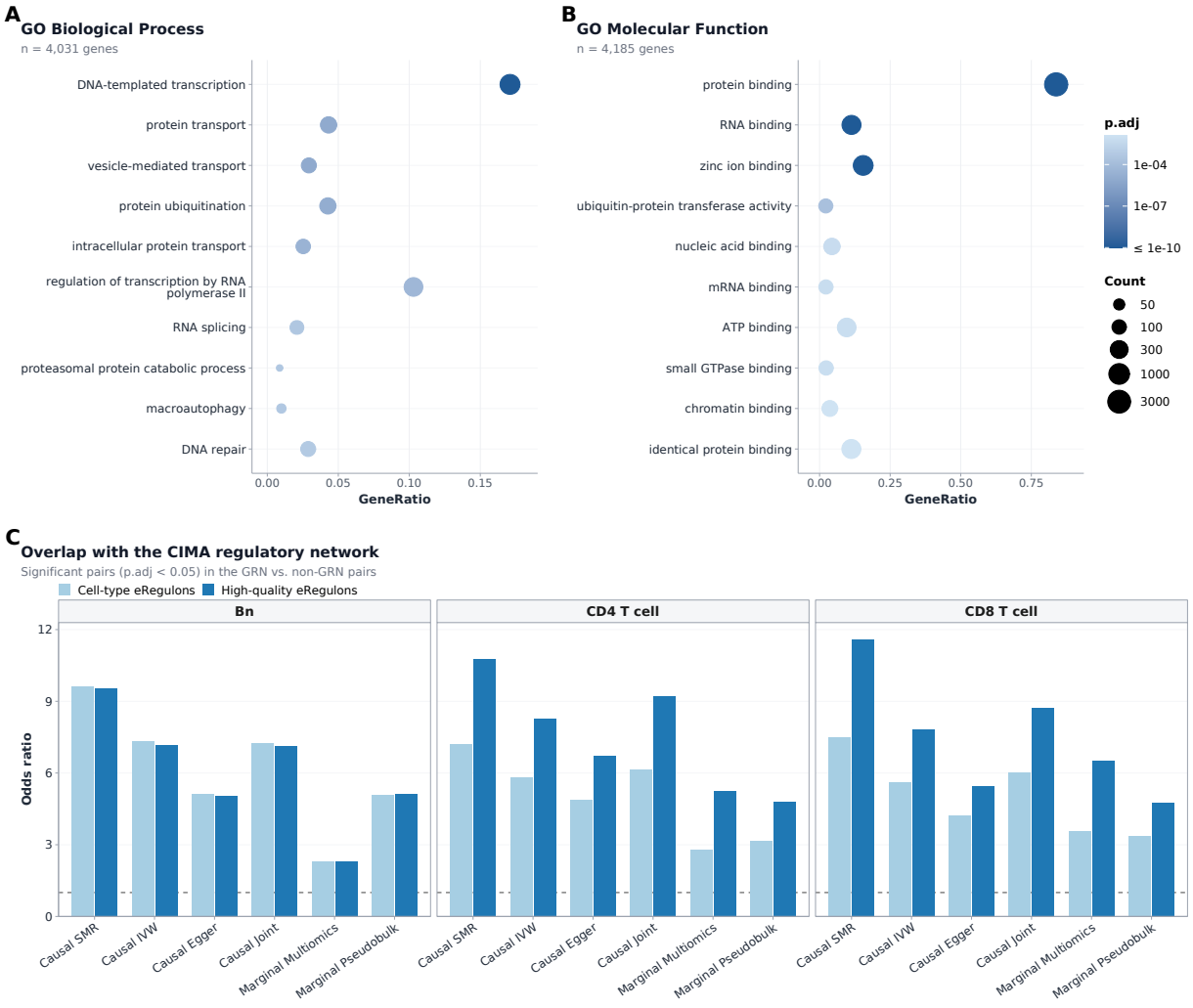


Fig 3. GO enrichment summaries and overlap with the CIMA regulatory network. **A,** Biological Process terms, with 4,031 genes reported in the panel. **B,** Molecular Function terms, with 4,185 genes reported in the panel. The horizontal axis gives GeneRatio, bubble size represents the displayed term count, and color represents adjusted p-values. **C,** Odds ratio for a link significant for each method (within-gene BH-adjusted $p < 0.05$, a stricter threshold than the $q < 0.1$ used elsewhere) being present in the CIMA SCENIC+ regulatory network, compared with links not in the network (2×2 table; all tested pairs in the cell type as background). Light bars use eRegulons active in the cell type (cell-type eRegulons), and dark bars the subset of these that belong to the high-quality eRegulons of the corresponding lineage (high-quality eRegulons); these contained 864, 2,520, and 1,364 tested pairs in Bn, CD4 T cells, and CD8 T cells (cell-type eRegulons) and 817, 458, and 382 (high-quality eRegulons). Fisher's exact test $P < 10^{-30}$ for every bar; the dashed line marks an odds ratio of 1. "Causal Joint" denotes CCT.

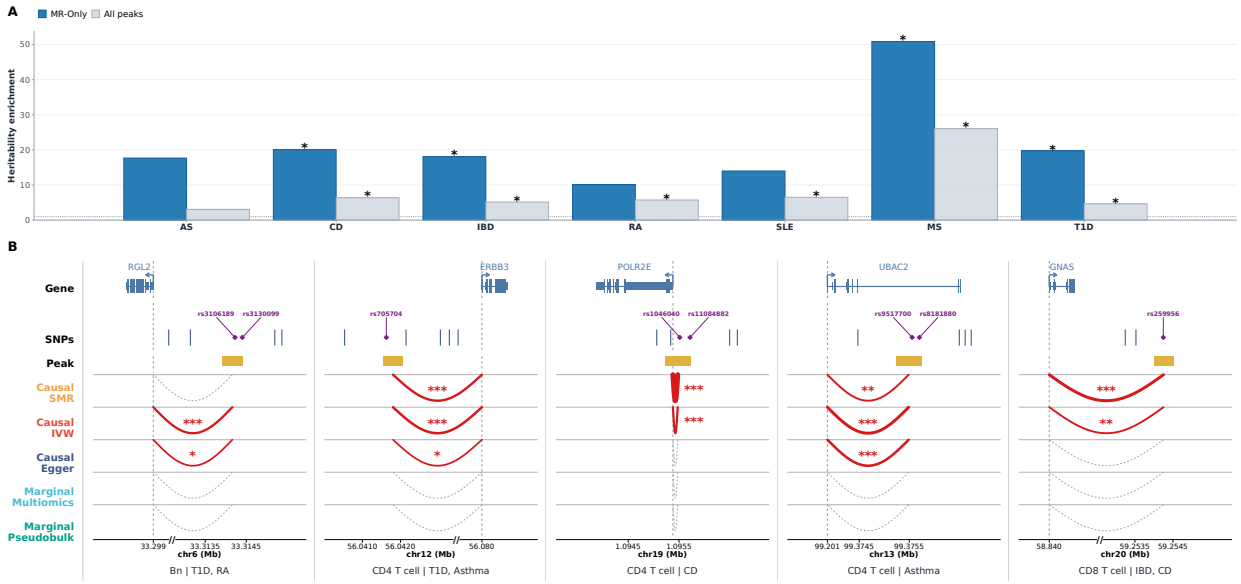


Fig 4. Disease-heritability enrichment and GWAS-overlapping regulatory links. **A** Heritability enrichment of the combined MR-only and all-candidate-peak S-LDSC annotations for each GWAS trait, shown on a common y axis; asterisks mark the annotated enrichment results (p value < 0.1). **B** Tracks show the gene, MR instruments (dark-blue ticks), GWAS variants within the peak (purple diamonds), the peak (gold bar), and one link per method (Causal SMR, IVW, and Egger; Marginal Multiomics and Pseudobulk). Significant links are drawn in red, and line width increases with $-\log_{10}$ adjusted p; gray dashed links are nonsignificant. Link stars denote adjusted $p < 0.05$ (*), 0.01 (**), and 0.001 (***).

Discussion

Preliminary outline only.

- **Principal findings.**
- **Complementary biological information.**
- **Limitations and future directions.**

There is one limit of our study. MR usually study Two sample data, where the variants-exposure and variants-outcome comes from two independent study. The two sample setting is important to get the causal effects. However, in our study, caQTL and eQTL data come from the same population. This might introduce some bias towards the true causal effects. We did sensitivity analysis by selecting SNPs under the null hypothesis and checked the correlations of residuals (Figure S1). Both of global and peak-gene pair-specific correlation support that there is little correlations between the two marginal associations in our study. The boxplots show the p values and correlation estimates for different number of SNPs, where more SNPs in one

peak-gene pairs indicate the clearer independence of residuals. The correlation between two residuals might be negligible. In the further study, we can also borrow eQTL data and caQTL from two separately study to eliminate the poential bias.

Another issue we want to discuss is the orientation of the genetics variants. Association of genetics variants can be quoted with respecrt to either the major or the minor allele. Causal IVW, which lack of intecept in the regression, provide exactly the same estimates using different of orientation of alleles (Figure S2). However, causal Egger results would be different. we follow the common setting of MR Egger (reference), and orientate the genetic variants so that the association with the risk factor all have the positive sign. This means that the directional pleitropy is define with respect to the risk factor-increasing allele. (Figure S2)

Materials and Methods

Data sources

Our study was conducted on the the Chinese Immune Multi-Omics Atlas (CIMA) (OMIX009786), which provided cell-type-specific caQTL and eQTL summary statistics, and single-cell RNA and ATAC count (Citation). We also used another external PBMC multiome data for the marginal multiome method.

CIMA provided hierarchical cell-type annotations, and we chose to focus on three major naive lymphocyte populations: Bn_TCL1A (naive B cells, marker gene *TCL1A*), CD4_Tn_CCR7 (naive $CD4^+$ T cells, marker gene *CCR7*), and CD8_Tn_CCR7 (naive $CD8^+$ T cells, marker gene *CCR7*). We will use Bn, CD4 T cells, and CD8 T cells throughout the paper for simplicity.

External multiome data for WNN

Data Preprocessing

- **caQTL filteration**
- **LD Clumping; allele must be mateched**
- **LD matrix calculation from PLINK**
- **Marginal Analysis, WNN, the external PBMC information**
- **Marginal Analysis, Pseudobulk, scRNA/snATAC matrix QC**
- **Candidate Peak-Gene Pairs**

CIMA conducted eQTL study for each gene and all SNPs within 1MB of the TSS of the genes. Similarly,

caQTL study was performed for each detected chromatin accessibility peaks and all SNPs within 1MB of the midpoint of the peaks. We choose to study all genes provided by CIMA eQTL study and all chromatin accessibility peaks which are 1MB within the TSS of the genes. In other words, we choose to study the association of all peak-gene pairs whose distance was less or equal to 1MB on the genome. However, several filtering steps were necessary. First, because under the MR framework, instrument variables (IVs) should be associated with the exposure. For each cell type, we only selected significant caQTLs (nominal p value $<$ 1e-8). Second, for each peak-gene pairs, we deleted duplicate instrumental variables. That is, there might be some SNPs as instrumental variables which has identical or highly samilar caQTL and eQTL. For those SNPs with the highly samilar caQTL and eQTL data (difference of both results are $<$ 1e-3), we randomly keep one SNP. Third, we dropped the chromatin accessibility peaks with less than three significant caQTLs (nominal p value $<$ 1e-8), since we needed sufficient instrument variables to fit the causal IVW and causal MR Egger models discussed below. Fourth, we only selected the peak-gene pairs in which both cell-type specific eQTL and caQTL data were available, and the pseudobulk ATAC and gene expression matrix were available, and the external PBMC data were also available. In the end, we got XXX peak-gene pairs for Bn, XXX peak-gene pairs for CD4 T cells, and XXX peak-gene pairs for CD8 T cells. (we need to move this part to the end, how to construct peak-gene pairs.)

LD clumping: We did more steps on caQTL data. After we select significant caQTL data (nominal caQTL p value $<$ 1e-8), we needed to perform LD clumping. We did the LD clumping for each peak-gene pairs separately using PLINK (version) with 'clump_r2=0.99', 'clump_kb =200000'. Variants were matched to the hg38 East Asian reference panel (version). The LD clumping step was not used to select independent SNPs, but to avoid perfect collinearity among highly correlated instrument variables.

LD matrix calculation: Since we implemented the extended version of IVW and MR-Egger models with correlated instrument variable, we needed to calculate the covariance matrix. The covariance matrix was obtained by the PLINK (version) with parameter –keep-allele-order, –r square, –write-snplist. PLINK would calculate the pairwise correlations between genotype allele counts based on the same reference panel (hg38 East Asian, version). The parameter 'keep-allele-order' was important to keep the effect allele of instrument variables and reference genome the same during LD matrix calculation. Moreover, because MR-Egger assumed the SNP-exposure effect were all positive (supplementary files, method section), we would orientate the genetics variants so that all caQTL effects were positive. The LD matrix was reoriented consistently with the exposure and outcome effects. In the end, we get the LD matrix.

Get Pseudobulk matrix: We directly used the cell-type specific scRNA and scATAC matrix provied by CIMA paper, and calculated the number of cells per individual. The individuals meeting the upper cutoff and the

lower cutoff criteria in both two modalities could be retained. Specifically, the lower cutoff was at least 100 cells for each individuals, and the upper cutoff was the median cell numbers of individuals plus three scaled median absolute deviations. After filtering out unqualified individuals, we got 335 individuals in Bn, 372 individuals in CD4 T cells, 346 individual in CD8 T cells. We separately summed gene expression and peak accessibility. We further performed CPM normalization and log1p transformation. Finnaly, we get cell-type specific, individual level pseudobulk matrix used for the marginal pseudobulk method.

WNN meta cells: Add how to get the ATAC_RNA_WNN.rda later. We used a previously processed PBMC multiome dataset, and the ATAC fragment accounts were requantified over CIMA-defined peak intervals by Signac (version) CreateFragmentObject function. The CIMA-define peaks were all peaks available in the caQTL data provided by CIMA. Then we conducted the weighted nearest-neighbor analysis using RNA principal components 1–50 and ATAC LSI dimensions 2–50 and get new WNN metacells. We summed ATAC counts and RNA expression across cells within each WNN cluster, and normalized to one million total counts per cluster. We get 80 metacells.

MR Framework

Mendelian randomization (MR) uses genetic variants as instrumental variables (IVs) to investigate the causal effect of an exposure on an outcome. In this study, we applied MR to investigate the effects of chromatin accessibility on gene expression for candidate peak–gene pairs. For each pair, the exposure X was chromatin accessibility at the selected ATAC-seq peak, and the outcome Y was expression of the corresponding gene.

A SNP is a valid instrument if it satisfies three assumptions:

1. **Relevance:** the SNP is associated with the exposure, here chromatin accessibility at the selected peak.
2. **Independence:** the SNP is independent of confounders of the exposure–outcome relationship, conditional on the covariates used in the analysis.
3. **Exclusion restriction:** the SNP affects the outcome only through the exposure.

The exclusion restriction can be violated by horizontal pleiotropy, which occurs when a SNP affects gene expression through pathways other than accessibility at the selected peak. IVW and MR-Egger differ in how they accommodate such effects. IVW provides consistent estimation when instruments are valid and can also accommodate balanced pleiotropy under additional assumptions. MR-Egger allows for directional pleiotropy under the Instrument Strength Independent of Direct Effect (InSIDE) assumption, as described below.

To connect individual-level associations with the summary statistics used in MR, consider a given peak–gene pair with J candidate instruments. Let $G_{ij} \in \{0, 1, 2\}$ denote the number of copies of the effect allele at SNP

j in individual i, where $j = 1, \ldots, J$ and $i = 1, \ldots, N$. Let X_i and Y_i denote the corresponding accessibility and expression measurements. For each SNP separately, the marginal associations can be represented as

$$X_i = \mu_{Xj} + \gamma_j G_{ij} + \varepsilon_{ij}^X,$$
$$Y_i = \mu_{Yj} + \Gamma_j G_{ij} + \varepsilon_{ij}^Y,$$

where μ_{Xj} and μ_{Yj} are intercepts, and γ_j and Γ_j are the population-level SNP–exposure and SNP–outcome associations, respectively. Covariate terms are omitted for simplicity.

We performed MR using caQTL and eQTL summary statistics. For SNP j, these comprise the estimated SNP–exposure association $\hat{\gamma}_j$ with standard error s_{Xj}, and the estimated SNP–outcome association $\hat{\Gamma}_j$ with standard error s_{Yj}. Both associations were aligned to the same effect allele.

Under a linear model with a common causal effect β, the population-level associations satisfy

$$\Gamma_j = \beta\gamma_j + \alpha_j, \tag{1}$$

where $\beta\gamma_j$ is the component mediated through accessibility at the selected peak, and α_j is the component not mediated through that exposure. For a valid instrument, $\alpha_j = 0$, and the ratio Γ_j/γ_j identifies β. This relationship forms the basis of the IVW and MR-Egger methods introduced in the following sections.

Inverse-variance weighted method

For uncorrelated genetic variants, the inverse-variance weighted (IVW) method combines SNP-specific ratio estimates to estimate the exposure–outcome effect. We use the NO Measurement Error (NOME) assumption that uncertainty in the estimated SNP–exposure associations ($\hat{\gamma}_j$) is negligible. Conditional on the exposure-association estimates, IVW is formulated as a weighted linear regression through the origin [1,2]:

$$\hat{\Gamma}_j = \beta\hat{\gamma}_j + \varepsilon_j, \qquad \varepsilon_j \overset{\text{ind}}{\sim} \mathcal{N}(0, \phi s_{Yj}^2), \quad j = 1, \ldots, J, \tag{2}$$

where s_{Yj} is the standard error of $\hat{\Gamma}_j$ and ϕ is a residual variance multiplier. The fixed-effect model sets $\phi = 1$, whereas the multiplicative random-effects model estimates ϕ for overdispersion. IVW can model balanced pleiotropy with $E(\alpha_j \mid \gamma_j) = 0$ [2].

The IVW estimate minimizes the weighted residual sum of squares

$$Q(\beta) = \sum_{j=1}^{J} \frac{(\hat{\Gamma}_j - \beta\hat{\gamma}_j)^2}{s_{Yj}^2}, \tag{3}$$

giving

$$\hat{\beta}_{\mathrm{IVW}} = \frac{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j\hat{\Gamma}_j}{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j^2} = \frac{\sum_{j=1}^{J} w_j\hat{\beta}_j}{\sum_{j=1}^{J} w_j}, \tag{4}$$

where $\hat{\beta}_j = \hat{\Gamma}_j/\hat{\gamma}_j$ and $w_j = \hat{\gamma}_j^2/s_{Yj}^2$. Thus, IVW is an average of the individual ratio estimates weighted by the inverse variances [1].

The conditional variance is

$$\mathrm{Var}(\hat{\beta}_{\mathrm{IVW}} \mid \hat{\boldsymbol{\gamma}}) = \frac{\phi}{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j^2}, \tag{5}$$

where $\hat{\boldsymbol{\gamma}}$ is the vector of estimated SNP–exposure associations. Residual dispersion is estimated by

$$\hat{\phi} = \frac{Q(\hat{\beta}_{\mathrm{IVW}})}{J-1}. \tag{6}$$

We choose the random-effects analysis to handle the overdispersion should $\tilde{\phi} = \max(1, \hat{\phi})$ [1]. The resulting standard error can be formally written as:

$$\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{IVW}}) = \left\{ \frac{\tilde{\phi}}{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j^2} \right\}^{1/2}. \tag{7}$$

We do the inference by:

$$T_{\mathrm{IVW}} = \frac{\hat{\beta}_{\mathrm{IVW}}}{\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{IVW}})} \tag{8}$$

with a $t_{J-1}(0)$ reference distribution.

MR-Egger regression

MR-Egger extends the zero-intercept IVW regression by estimating an intercept, thereby allowing for a nonzero average pleiotropic effect. For uncorrelated SNPs, and treating the estimated SNP–exposure associations as fixed under the NOME approximation, the working model is

$$\hat{\Gamma}_j = \alpha_0 + \beta\hat{\gamma}_j + \varepsilon_{\mathrm{E},j}, \qquad \varepsilon_{\mathrm{E},j} \overset{\text{ind}}{\sim} \mathcal{N}(0, \phi_{\mathrm{E}} s_{Yj}^2), \tag{9}$$

where α_0 is the intercept and ϕ_{E} is a residual variance multiplier [1]. The intercept α_0 summarizes average pleiotropy under the identifying assumptions. Both exposure and outcome associations are oriented to the exposure-increasing allele, so that $\hat{\gamma}_j > 0$.

Causal interpretation of the slope requires the Instrument Strength Independent of Direct Effect (InSIDE) assumption: the true SNP–exposure associations γ_j are independent of the pleiotropic effects α_j. InSIDE allows MR-Egger to account for directional pleiotropy [1, 2].

Let $v_j = s_{Yj}^{-2}$ denote the regression weights and define

$$\bar{\gamma}_v = \frac{\sum_{j=1}^J v_j\hat{\gamma}_j}{\sum_{j=1}^J v_j}, \qquad \bar{\Gamma}_v = \frac{\sum_{j=1}^J v_j\hat{\Gamma}_j}{\sum_{j=1}^J v_j}. \tag{10}$$

The weighted least-squares estimates minimize

$$Q_{\mathrm{E}}(\alpha_0, \beta) = \sum_{j=1}^J v_j(\hat{\Gamma}_j - \alpha_0 - \beta\hat{\gamma}_j)^2. \tag{11}$$

Writing $S_{\gamma\gamma} = \sum_{j=1}^J v_j(\hat{\gamma}_j - \bar{\gamma}_v)^2$, the estimates are:

$$\begin{aligned} \hat{\beta}_{\mathrm{Egger}} &= \frac{\sum_{j=1}^J v_j(\hat{\gamma}_j - \bar{\gamma}_v)(\hat{\Gamma}_j - \bar{\Gamma}_v)}{S_{\gamma\gamma}}, \\ \hat{\alpha}_0 &= \bar{\Gamma}_v - \hat{\beta}_{\mathrm{E}}\bar{\gamma}_v. \end{aligned} \tag{12}$$

Under the working model, the conditional slope variance is

$$\mathrm{Var}(\hat{\beta}_{\mathrm{Egger}} \mid \hat{\boldsymbol{\gamma}}) = \frac{\phi_{\mathrm{Egger}}}{S_{\gamma\gamma}}. \tag{13}$$

Thus, slope precision depends on variation in the SNP–exposure associations, rather than their overall magnitude alone. Similar exposure associations across SNPs can yield an imprecise MR-Egger estimate even when the individual associations are strong [1].

Because the regression estimates both an intercept and a slope, residual dispersion is estimated using $J - 2$ degrees of freedom:

$$\hat{\phi}_{\mathrm{Egger}} = \frac{Q_{\mathrm{E}}(\hat{\alpha}_0, \hat{\beta}_{\mathrm{E}})}{J - 2}. \tag{14}$$

For $J > 2$ and $S_{\gamma\gamma} > 0$, we use $\tilde{\phi}_{\mathrm{E}} = \max(1, \hat{\phi}_{\mathrm{E}})$ to prevent underdispersion. The slope and intercept standard errors are

$$\begin{aligned} \widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{Egger}}) &= \left(\frac{\tilde{\phi}_{\mathrm{Egger}}}{S_{\gamma\gamma}} \right)^{1/2}, \\ \widehat{\mathrm{SE}}(\hat{\alpha}_0) &= \left\{ \tilde{\phi}_{\mathrm{Egger}} \left[\frac{1}{\sum_{j=1}^{J} v_j} + \frac{\bar{\gamma}_v^2}{S_{\gamma\gamma}} \right] \right\}^{1/2}. \end{aligned} \tag{15}$$

The slope test evaluates $H_0 : \beta = 0$ using

$$T_{\mathrm{Egger}} = \frac{\hat{\beta}_{\mathrm{Egger}}}{\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{Egger}})}. \tag{16}$$

Two-sided p-values use a $t_{J-2}(0)$ reference distribution

GLS-IVW with correlated variants

SNPs within *cis*-regulatory regions are often correlated due to linkage disequilibrium (LD). Stringent LD clumping to obtain independent instruments can yield too few variants per peak–gene pair, thereby limiting statistical power. We apply the generalized least squares (GLS) extension of the IVW method to account for remaining LD among instruments after clumping [1,3]. The final clumping threshold is pending reconciliation in the Data Preprocessing subsection.

For a given peak–gene pair, define the estimated SNP-exposure, SNP-outcome assocaition as:

$$\hat{\boldsymbol{\gamma}} = (\hat{\gamma}_1, \ldots, \hat{\gamma}_J)^{\mathsf{T}}, \qquad \hat{\boldsymbol{\Gamma}} = (\hat{\Gamma}_1, \ldots, \hat{\Gamma}_J)^{\mathsf{T}}. \tag{17}$$

Let $\boldsymbol{R}$ be the genotype-correlation matrix estimated from the 1000 Genomes East Asian reference panel, and let $\boldsymbol{D}_Y = \mathrm{diag}(s_{Y1}, \ldots, s_{YJ})$, the standard deviation of SNP-outcome association. Using an LD-based approximation to the sampling covariance of the outcome-association estimates, we construct:

$$\boldsymbol{\Sigma} = \boldsymbol{D}_Y \boldsymbol{R} \boldsymbol{D}_Y, \tag{18}$$

The IVW model turns:

$$\hat{\boldsymbol{\Gamma}} = \beta \hat{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}_J(\mathbf{0}, \phi \boldsymbol{\Sigma}), \tag{19}$$

where ϕ is a residual variance multiplier under random effect model. We need to minimize the object

function:

$$Q_{\mathrm{GLS}}(\beta) = (\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}}) \tag{20}$$

gives

$$\hat{\beta}_{\text{GLS-IVW}} = \frac{\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{\Gamma}}}{\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{\gamma}}}. \tag{21}$$

This is the correlated-variant extension of the IVW estimator [1,3].

Strong LD can make $\boldsymbol{\Sigma}$ nearly singular, so that direct inversion becomes numerically unstable. We therefore regularize the covariance weighting using a truncated eigendecomposition:

$$\boldsymbol{\Sigma} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}}, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_J), \tag{22}$$

where $\boldsymbol{V}$ is orthogonal and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_J$. Let $\lambda_\ell^+ = \max(\lambda_\ell, 0)$ and define

$$m_\eta = \min\left\{ m \in \{1, \ldots, J\} : \frac{\sum_{\ell=1}^{m} \lambda_\ell^+}{\sum_{\ell=1}^{J} \lambda_\ell^+} \geq \eta \right\}, \qquad \eta = 0.999. \tag{23}$$

The retained directions are

$$\mathcal{K} = \left\{ \ell : 1 \leq \ell \leq m_\eta,\ \lambda_\ell > 10^{-10}\lambda_1 \right\}, \qquad k = |\mathcal{K}|. \tag{24}$$

The relative eigenvalue threshold excludes directions with very small or non-positive eigenvalues.

We denote the truncated covariance approximation by

$$\widetilde{\boldsymbol{\Sigma}} = \boldsymbol{V}\,\mathrm{diag}(\widetilde{\lambda}_1, \ldots, \widetilde{\lambda}_J)\boldsymbol{V}^{\mathsf{T}}, \qquad \widetilde{\lambda}_\ell = \begin{cases} \lambda_\ell, & \ell \in \mathcal{K}, \\ 0, & \ell \notin \mathcal{K}. \end{cases} \tag{25}$$

When $k < J$, this matrix is singular. We therefore use its Moore–Penrose pseudoinverse,

$$\widetilde{\boldsymbol{\Sigma}}^{+} = \boldsymbol{V}\,\mathrm{diag}(d_1, \ldots, d_J)\boldsymbol{V}^{\mathsf{T}}, \qquad d_\ell = \begin{cases} \lambda_\ell^{-1}, & \ell \in \mathcal{K}, \\ 0, & \ell \notin \mathcal{K}, \end{cases} \tag{26}$$

which inverts the covariance on the retained eigenspace and assigns zero weight to discarded directions.

The regularized GLS-IVW estimator minimizes

$$Q_{\mathcal{K}}(\beta) = (\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}}), \tag{27}$$

giving

$$\hat{\beta}_{\text{GLS-IVW}} = \frac{\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\Gamma}}}{\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\gamma}}}. \tag{28}$$

Provided that the denominator is positive, its conditional variance under the working covariance model is

$$\operatorname{Var}\left(\hat{\beta}_{\text{GLS-IVW}} \mid \hat{\boldsymbol{\gamma}}, \boldsymbol{\Sigma}\right) = \frac{\phi}{\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\gamma}}}. \tag{29}$$

GLS MR-Egger with correlated variants

GLS MR-Egger extends the correlated-variant IVW model by estimating an intercept [1]. Using the association vectors and LD-based covariance matrix defined above, the MR Egger formula is

$$\hat{\boldsymbol{\Gamma}} = \alpha_0\mathbf{1}_J + \beta\hat{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}_{\text{Egger}}, \qquad \boldsymbol{\varepsilon}_{\text{Egger}} \sim \mathcal{N}_J(\mathbf{0}, \phi_{\text{Egger}}\boldsymbol{\Sigma}), \tag{30}$$

where $\mathbf{1}_J$ is a vector of ones, α_0 is the intercept, and ϕ_{Egger} is a residual variance multiplier. The estimated SNP–exposure associations $\hat{\gamma}$ are treated as fixed under the NOME approximation.

To stabilize covariance weighting, we use the truncated covariance matrix $\widetilde{\boldsymbol{\Sigma}}$ and its Moore–Penrose pseudoinverse $\widetilde{\boldsymbol{\Sigma}}^{+}$ defined for GLS-IVW [Eq. (26)]. Define

$$\boldsymbol{X} = \begin{bmatrix} \mathbf{1}_J & \hat{\boldsymbol{\gamma}} \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{pmatrix} \alpha_0 \\ \beta \end{pmatrix}. \tag{31}$$

The coefficients minimize

$$Q_{\text{Egger}}(\boldsymbol{\theta}) = (\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\boldsymbol{\theta})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\boldsymbol{\theta}). \tag{32}$$

Provided that $\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\boldsymbol{X}$ is nonsingular, the estimates are

$$\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}} = \begin{pmatrix} \hat{\alpha}_0 \\ \hat{\beta}_{\text{GLS-Egger}} \end{pmatrix} = \left(\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\boldsymbol{X}\right)^{-1}\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\Gamma}}. \tag{33}$$

Under the working covariance model, their conditional covariance matrix is

$$\mathrm{Var}\left(\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}} \mid \hat{\boldsymbol{\gamma}}, \boldsymbol{\Sigma}\right) = \phi_{\text{Egger}} \left(\boldsymbol{X}^{\mathsf{T}} \widetilde{\boldsymbol{\Sigma}}^{+} \boldsymbol{X}\right)^{-1}. \tag{34}$$

The slope standard error is obtained from the second diagonal element, using the dispersion adjustment $\widetilde{\phi}_{\text{Egger}} = \max(1, \hat{\phi}_{\text{Egger}})$. The fitted residual quadratic form is $Q_{\text{Egger}}(\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}})$.

(We can explain the filteration rule here, after show MR-Egger and IVW methods.) Pairs with any missing p-value are removed, then pairs with IVW residual variance greater than 10 are removed. BH adjustment is applied separately within each gene and method [4]. Discovery uses $q < 0.1$.

Summary-data-based Mendelian randomization

Summary-data-based Mendelian randomization (SMR) uses the associations of a single genetic instrument with an exposure and an outcome to estimate their relationship from summary statistics [5]. We applied this approach to chromatin accessibility and gene expression. For each peak–gene pair, we selected the SNP with the smallest nominal caQTL p-value among the variants retained after the filtering described above, breaking ties by the largest absolute caQTL effect-to-standard-error ratio. Denoting this SNP by $*$, the effect estimate is

$$\hat{\beta}_{\text{SMR}} = \frac{\hat{\Gamma}_*}{\hat{\gamma}_*}. \tag{35}$$

The SMR test uses a delta-method approximation to the variance of this ratio, accounting for uncertainty in both associations [5]. In our implementation, the sampling covariance between the caQTL and eQTL effect estimates was omitted from this approximation.

For the selected SNP, we converted the two-sided nominal caQTL and eQTL p-values, p_X and p_Y, to nonnegative normal-equivalent statistics:

$$\begin{aligned} z_X &= \Phi^{-1}(1 - \tilde{p}_X/2), \qquad z_Y = \Phi^{-1}(1 - \tilde{p}_Y/2), \\ \tilde{p}_X &= \max(p_X, 10^{-300}), \quad \tilde{p}_Y = \max(p_Y, 10^{-300}), \end{aligned} \tag{36}$$

where Φ is the standard normal cumulative distribution function. The numerical implementation evaluates inverse upper-tail probabilities. Using these p-derived statistics, we calculated

$$\begin{aligned} \widehat{\mathrm{SE}}(\hat{\beta}_{\text{SMR}}) &= |\hat{\beta}_{\text{SMR}}| \sqrt{z_X^{-2} + z_Y^{-2}}, \\ T_{\text{SMR}} &= \frac{z_X^2 z_Y^2}{z_X^2 + z_Y^2}. \end{aligned} \tag{37}$$

The SMR p-value was the upper-tail probability of T_{SMR} under an approximate χ^2_1 reference distribution. The effect direction was determined by the aligned association estimates in the ratio. We did not perform the HEIDI heterogeneity test; SMR significance alone does not distinguish mediation from horizontal pleiotropy or linkage between distinct causal variants [5].

Marginal pseudobulk correlations

We evaluated marginal peak–gene associations using pseudobulk RNA and ATAC profiles within each cell type. Following the quality control described in Data Preprocessing, counts were summed across cells from the same individual within each cell type to construct gene-by-individual and peak-by-individual pseudobulk matrices. We retained individuals with both RNA and ATAC profiles, yielding 335 individuals for Bn, 372 for CD4 T cells, and 346 for CD8 T cells. For each modality, counts were normalized to counts per million (CPM) using the corresponding individual-level library size and transformed using $\log(1 + \mathrm{CPM})$. Library sizes were calculated across all input genes or peaks.

For each candidate peak–gene pair, we calculated Spearman's rank correlation between peak accessibility and gene expression across matched individuals:

$$\widehat{\rho}_{pg}^{\mathrm{PB}} = \mathrm{Cor}_{\mathrm{S}}\left(\boldsymbol{A}_p, \boldsymbol{E}_g\right), \tag{38}$$

where $\boldsymbol{A}_p$ and $\boldsymbol{E}_g$ are the log-transformed CPM vectors for peak p and gene g, respectively, across matched individuals, and $\mathrm{Cor}_{\mathrm{S}}$ denotes Spearman's rank correlation. Two-sided approximate p-values were obtained using R's `cor.test` with `method="spearman"` and `exact=FALSE`. No covariate residualization was applied at the correlation stage. P-values were adjusted using the Benjamini–Hochberg procedure within each gene, and peak–gene pairs with FDR ≤ 0.1 were considered significant.

WNN metacell correlations

We evaluated marginal peak–gene associations using an external PBMC multiome dataset with 80 precomputed weighted nearest-neighbor (WNN) metacell groups. The same RNA and ATAC metacell profiles were used for all three CIMA cell-type analyses, without restricting metacells to the corresponding cell type. WNN defined the cell groups, whereas peak–gene associations were quantified by correlations across metacells.

TODO: Add the external dataset description and citation, the WNN method citation, and details of the original metacell construction and RNA aggregation and normalization.

To align accessibility features with CIMA, external ATAC fragments were recounted over CIMA peak intervals in hg38 using Signac's `FeatureMatrix`, after excluding intervals that overlapped other peaks in the input set. ATAC counts were summed within each metacell and normalized to counts per million using the total counts across retained peaks in that metacell. The supplied RNA metacell matrix and group assignments were retained. A $\log(1+x)$ transformation was then applied to both the normalized ATAC matrix and the supplied RNA matrix.

For each candidate peak–gene pair, we calculated Spearman's rank correlation between peak accessibility and gene expression across the 80 metacells:

$$\widehat{\rho}_{pg}^{\mathrm{WNN}} = \mathrm{Cor}_{\mathrm{S}}\left(\boldsymbol{A}_p^{\mathrm{MC}}, \boldsymbol{E}_g^{\mathrm{MC}}\right), \tag{39}$$

where $\boldsymbol{A}_p^{\mathrm{MC}}$ and $\boldsymbol{E}_g^{\mathrm{MC}}$ denote the transformed accessibility and expression vectors for peak p and gene g, respectively, across metacells, and $\mathrm{Cor}_{\mathrm{S}}$ denotes Spearman's rank correlation. Metacells served as the units of observation. Two-sided approximate p-values were calculated as described in Marginal pseudobulk correlations. No covariate residualization, cell-type adjustment, or weighting by metacell size was applied. The resulting correlations therefore include variation across PBMC cell types and states. P-values were adjusted using the Benjamini–Hochberg procedure within each gene, separately for each CIMA cell-type analysis, and peak–gene pairs with FDR ≤ 0.1 were considered significant.

Causal joint

The Cauchy combination test (CCT) is an efficient and powerful technique to combine multiple individual p-values, especially when the tests are correlated. Under the null hypothesis, for an individual p-value p_k, the transformed statistic follows a standard Cauchy distribution. We calculated the causal joint results by aggregating the three causal methods (IVW, MR-Egger, and SMR) using the Cauchy combination test with equal weights:

$$T = \frac{1}{3}\left\{\tan\left[\pi(0.5 - p_{\mathrm{IVW}})\right] + \tan\left[\pi(0.5 - p_{\mathrm{Egger}})\right] + \tan\left[\pi(0.5 - p_{\mathrm{SMR}})\right]\right\},$$

where p_{IVW}, p_{Egger}, and p_{SMR} denote the nominal p-values from GLS-IVW, GLS MR-Egger, and SMR, respectively. The causal joint results can be understood as the aggregated causal evidence for the studied peak–gene links.

Motif enrichment analysis

We implemented the motif enrichment analysis to determine whether the transcription factor binding motifs or functional sequence patterns are statistically overrepresented in the detected peaks compared to all background peaks. For each cell types (Bn, CD4 T cells, CD8 T cells) and each method (causal IVW, causal MR-Egger, causal SMR, marginal multiomics, and marginal pseudobulk), we separately selected significant peak-gene pairs (BH p.adjust < 0.1) and further extracted those peaks as the foreground set for the motif analysis. All available peaks provided by CIMA cell-type specific xQTL data were used as the background set. Motif occurrences were annotated with AddMotifs in Signac (citation, version) using the non-redundant JASPAR2020 CORE vertebrate collection (746 motifs) (citation) and the hg38 genome (BSgenome.Hsapiens.UCSC.hg38 [version]). We use the FindMotifs function in Signac to get the p value of motifs, and the p values were adjusted by the Benjamini–Hochberg (BH) procedure across motifs for each cell type and method. Motifs with adjusted $p < 0.05$ and fold enrichment > 1 were considered enriched. We were interested in the motif which were significant in at least one causal methods but were not significant by both marginal multiomics and marginal pseudobulk methods. In the table (table number), we should the most significant motifs only detected by causal methods for each cell types seperately.

Hypergeometric test

We carried out global hypergeometric tests to the existing enhancer and chromatin interaction databases: FANTOM5 (citation) linked enhancers and genes based on enhancer RNA expression; 4DGenome (citation) linked enhancers and genes based on physical interactions using chromatin-looping data; EnhancerAtlas (citation) reported enhancers using 12 high-throughput experimental methods; a list of cis-eQTLs within the whole blood mapped in European-American subjects was downlaoded from the GTEx Consortiums (citation); CHi-C captured contacts between gene promoters and distal regions (CHiCAGO score ≥ 5) (citation); H3K27ac HiChIP captures contacts among regions marked by active regulatory histone modification (FDR <0.1) (citation). We used liftover in R package rtracklayer (version) to map some genomics data in hg19 to hg38. We did global hypergeometric tests for each cell type and method seperately. Let k be the number of significantly linked peaks, q be the number of significantly linked peaks that overlap with annotations (e.g., annotated enhancers), m be the number of peaks that overlap with the annotations, and n be the number of peaks that do not overlap with annotations. The p-value of enrichment is derived from the hypergeometric distribution using the cumulative distribution function, coded as phyper(q-1, m, n, k, lower.tail=F) in R.

Gene ontology analysis

We performed GO enrichment analysis for Gene Ontology (GO) Biological Process and Gene Ontology (GO) Molecular Function terms using gene set enrichment analysis (GSEA). We collect genes from all significant peak-gene pairs significant in at least one causal method (IVW, MR Egger, SMR, p.adust < 0.1) not significant in both two marginal association method (marginal multiomics, marginal pseudobulk, p.adjust $>$ 0.1) from each cell type. We combine selected genes from three cell types. Because one gene could be tested with multiple peaks, each gene was scored by the -log10 of the smallest P value of all tested peak-gene links from different methods. GSEA was run with gseGO in the clusterProfiler package (citation, version), using gene-symbol annotations from org.Hs.eg.db (version), gene sets of 10–500 genes, a score exponent of 1, positive-only scoring (scoreType = "pos"), and BH adjustment of p-values.

Heritability enrichment analysis from S-LDSC

Heritability enrichment analysis was conducted using stratified linkage disequilibrium score (S-LDSC) regression (Version later) to study whether the peaks from the peak-gene pairs only detected by MR methods have higher enrichment score compared to background peaks. We collected all peak-gene pairs which are significant in at least one causal method (IVW, MR Egger, SMR, p.adust < 0.1) but not significant in both two marginal association method (marginal multiomics, marginal pseudobulk, p.adjust > 0.1) across three cell types. Background peaks are peaks from all available peak-gene pairs provided by CIMA QTL data (reference), CIMA conduct QTL analysis for all peaks within 1MB of genes. S-LDSC calculate the heritability enrichment score by the ratio of the heritability explained versus the proportion of SNPs, based on GWAS summary data. We assessed heritability enrichment using S-LDSC with the baselineLD v2.2 annotations and 1000 Genomes Phase 3 European LD reference data, restricting regression to HapMap3 SNPs and using the standard regression weights that exclude the MHC region. Nine GWAS summary statistics used in the study were: ankylosing spondylitis(AS), crohn's disease(CD), inflammatory bowel disease(IBD), rheumatoid arthritis(RA), systemic lupus erythematosus(SLE), multiple sclerosis(MS), Type 1 diabetes(T1B), Asthma, and Primary biliary cirrhosis (PBC). GWAS summary statistics for these dieases and traits were obtained from `https://zenodo.org/records/10515792/files/sumstats_indep107.tgz?download=1` (CD, IBD, RA, SLE, MS, T1B, Asthma, PBC), Pan-UK Biobank `https://pan.ukbb.broadinstitute.org/` (AS), and MS link (Later). For dataset whose genomic location are in hg19, we use liftover in R package rtracklayer (version) to map the location to hg38.

Sensitivity analysis: Overlapping samples

Let's recall the two fundamental formula of caQTL and eQTL study, which will be the exposure and outcome in the MR framework. For each SNP separately, the marginal associations can be represented as

$$X_i = \mu_{Xj} + \gamma_j G_{ij} + \varepsilon_{ij}^X,$$
$$Y_i = \mu_{Yj} + \Gamma_j G_{ij} + \varepsilon_{ij}^Y,$$

where μ_{Xj} and μ_{Yj} are intercepts, and γ_j and Γ_j are the population-level SNP–exposure and SNP–outcome associations, respectively. MR study is usually base on Two-Sample data, where the SNP-exposure estimates and SNP-outcome estimates come from two independent but homogeneous samples. In this ways, the residuals ε_{ij}^X and ε_{ij}^Y are independent. However, in our study, caQTL and eQTL come from the same population, leading to some potential bias. In order to measures of the bias, we do the following sensitivity test for each of cell types separately.

We first do the LD pruning (–indep-pairwise 500kb 50 0.1, MAF >= 0.01) on the all available SNPs in the caQTL data. We further select the insignificant SNPs under the null (caQTL nominal p > 0.5, eQTL nominal p > 0.5). The marginal associations are under the null, there is no evidence to prove the marginal associations (γ_j and Γ_j) are nonzero values. Therefore, we can use the correlation of Z scores to approximate the correlation of residuals:

$$z_{Xj} = \frac{\hat{\gamma}_j}{s_{Xj}}, \qquad z_{Yj} = \frac{\hat{\Gamma}_j}{s_{Yj}}.$$

We calculate the whole-genome correlations of z_{Xj} and z_{Yj} for all SNPs for each cell types. And we also calculate the correlation of z_{Xj} and z_{Yj} for each peak-gene pairs with at least three SNPs after LD pruning. We show the box-plot of correlation and P values (Figure S1)

Acknowledgments

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Supporting information

All supporting figures and tables are provided in a separate file (`supplementary.pdf`).

S1 Fig. Null-SNP diagnostics in Bn, CD4 T cells, and CD8 T cells.

S2 Fig. MR-Egger estimates with and without allele orientation in Bn, CD4 T cells, and CD8 T cells.

S3 Fig. Overlap and effect directions of significant links in Bn and CD8 T cells.

S1 Table. Pooled hypergeometric enrichment p-values across regulatory references.

S2 Table. Pooled regulatory-reference fold enrichment across cell types and methods.

S3 Table. Data overview for the three cell types analyzed.